Overview

Today’s best coding agents rely on massive API models running in the cloud. What if every developer could rely on an autonomous agent equally capable offline, on consumer hardware? Post-train an open model into a reliable agent that navigates complex codebases and drafts fixes for real software issues, accelerating developer workflows on everyday hardware.

Start

9 days ago

Close

a month to go

Merger & Entry

Description

Note: there are two active competitions for the Gemma 4 Developer Agent Competition: the featured competition, and this paper track, where you can submit novel, unpublished research or document your approach to the prediction competition. Winning papers will be highlighted at a Google hosted event (such as a NeurIPS expo workshop) and awarded prizes.

Today’s most capable coding agents rely on internet-connected models running on powerful commercial infrastructure. That puts advanced software engineering out of reach for many developers.

Small local models have made rapid progress, but they still struggle with complex, multi-turn software engineering. Large codebases are hard to navigate, making reliable bug fixes and feature additions tough for models that fit on a single accelerator.

While The Gemma 4 Developer Agent Competition leaderboard ranks participants by bugs fixed, the Paper Track invites you to contribute research to advance the state-of-the-art in agentic software engineering. You will submit a Kaggle Writeup that describes your methods, experiments, and other details. Participation in the main competition isn't required to submit to the Paper Track, though we encourage participants to join both. Submissions can include any related research, as long as it is unpublished.

Your solution could help put powerful coding assistants in the hands of millions more developers. Sharing how you did it makes advanced coding tools accessible to everyone.

Timeline

- **September 22, 2026** - Start Date.
- **November 12, 2026** - Final Submission Deadline.

All deadlines are at 11:59 PM UTC on the corresponding day unless otherwise noted. The competition organizers reserve the right to update the contest timeline if they deem it necessary.

Evaluation

To be eligible to win the The Gemma 4 Developer Agent Paper Track Prize, you must join this competition and submit a Writeup that documents your research to advance the state-of-the-art for agentic software engineering.

Submissions for Paper Awards are evaluated equally across the following five criteria. Each criterion is scored on a scale from 0 (lowest) to 5 (highest), with the final score calculated as the average of all five.

Category	Description
Novelty	Does the work provide new insights, deepen understanding, or highlight important properties of existing methods?
Quality	How general and universal is the Submission approach beyond the competition? Does your submission translate to other similar problems? How well does your method generalize?
Relevance	Is there significance and/or potential impact of the research on the field of software engineering and agentic learning?
Verifiability	Does the paper include sufficient information to understand how an innovation works; to understand how data was obtained, analyzed, and interpreted?
Clarity	Is the paper clearly presented and written?

The Paper Award will be awarded to three Submissions with the most points. Your Paper rubric evaluation will not be shared with you. In the event of a tie, the Paper that was entered first to the Competition will be the winner. In the event a potential winner is disqualified for any reason, the Paper that received the next highest score rank will be chosen as the potential winner.

Submission Requirements

A valid submission must contain the following:

1. The Kaggle Writeup serves as your research paper project report. It should include:
- Title and subtitle
- Abstract
- Introduction
- Description of the research, including methods and experiments
- Related works and citations

(Optional) Public Notebook: If you choose to submit code, your code should be submitted as a public notebook in the Project Links field. Your notebook should be publicly accessible and not require a login or paywall. If you use a private Kaggle Notebook, it will automatically be made public after the deadline.

(Optional) Public Project Link: Optionally, instead of writing your paper in the Kaggle Writeup, you can also upload an arXiv-ready (LaTeX/Word) PDF version of your paper using the Public Project Link feature. It should be publicly accessible and not require a login or paywall.

To create a new Writeup, click on the "New Writeup" button here. After you have saved your Writeup, you should see a "Submit" button in the top right corner.

Submission Details

- **Scope**: 3,000 words maximum, reporting original, unpublished research.
- **Non-Archival**: Submission to this competition does not prevent submission to other conferences or venues.
- **Additional Assets**: The below assets are optional.

Submission Topics

Participants are encouraged to submit original, unpublished research submissions on topics including, but not limited to:

- **Tuning & Optimization**: Parameter-efficient fine-tuning and Reinforcement Learning (RL) approaches tailored to improve SWE agent capabilities.
- **Code Comprehension**: Novel techniques for code-graph generation, parsing, and embedding to enhance deep comprehension of complex repositories.
- **Tasks & Benchmarks**: Development of new tasks, evaluation datasets, or resources targeting code structure, graphs, and structured code generation.
- **Graph Reasoning**: Methods and architectures for improving Large Language Model (LLM) reasoning over large-scale graphs.

Share your writeup by November 12, 2026 at 11:59pm UTC. **Your final Submission must be made prior to the deadline. Any un-submitted or draft Writeups by the competition deadline will not be considered by the Judges**.

Note: If you attach a private Kaggle Resource to your public Kaggle Writeup, your private Resource will automatically be made public after the deadline.

Tracks and Awards

Total Prizes Available · $35,000

Prizes will be awarded amongst all submissions. Top submissions will be highlighted at a Google hosted expo during NeurIPS 2026. Prizes awarded for this competition are the following:



Overall Best Paper - $15,000

Overall best paper will be award the highest score calculated as the average of all five criteria.

$15,000



Best New Resource (Dataset, library, etc) - $10,000

A resource introduces a new dataset, tool, or software, for example. This is compared to an empirical paper which generates new knowledge/experiments. For examples of each, a new ‘dataset’ could leverage the provided embeddings or other sources to create a dataset that can be used for either evaluation of Gemma 4 or for fine-tuning it. A new ‘tool’ could be a library which uses Gemma and some other algorithm to cluster code based on underlying behavior. New ‘software’ could be a dashboard to explore Github repos that are similar in structure to a target repo.

$10,000



Best New Application - $10,000

A novel application is a new use-case (e.g. using code-graphs for writing TPU kernels). Show your creative uses of code-graphs and/or the provided embeddings to create novel use-cases. This may involve creating new evaluation datasets for entirely novel tasks. It could also involve application to existing benchmarks that differ from the provided ones, and using the code-graphs or embeddings where they have not been used before.

$10,000

Judges

Elan Markowitz

Machine Learning Engineer, Google

Citation

Elan Markowitz, Bryan Perozzi, Benedek Rózemberczki, Glenn Cameron, Hadi Hemmati, Yuchen Li, Michael Galkin, Majid Farhadi, Ryan Holbrook, and Ashley Oldacre. Google - The Gemma 4 Developer Agent Paper Track. https://www.kaggle.com/competitions/gemma-4-developer-agent-paper, 2026. Kaggle.